

Intelligent Crop Yield Prediction and Agentic Farm Advisory System

A decoupled full-stack machine learning architecture for precision agriculture and data-driven farm management

Project Overview



Yield Prediction

Supervised learning with ensemble models to forecast continuous crop yield from historical farm, soil, and weather data



Agentic Assistant

LangGraph-powered AI translates numerical predictions into structured, actionable agronomic recommendations



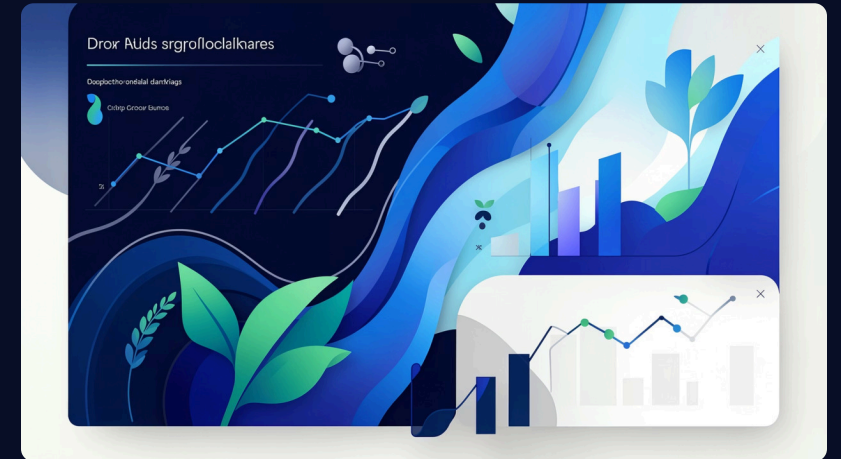
Production Architecture

Decoupled FastAPI backend and animated React frontend delivering scalable, secure agricultural analytics

Why Crop Yield Prediction Matters

Crop yield prediction is fundamental to agricultural planning and global food security. Our platform leverages advanced machine learning to forecast yield (hg/ha) based on critical inputs including average rainfall, temperature patterns, pesticide usage, and crop type classification.

By transforming raw data into actionable intelligence, farmers and agricultural planners can make informed decisions about resource allocation, planting strategies, and risk mitigation.



System Architecture

1

Frontend Layer

React/Vite SPA with Tailwind CSS and Framer Motion animations

2

API Interface

Secure HTTPS endpoints with JWT authentication

3

Backend Engine

FastAPI Python server handling ML inference and LLM integration

4

Data Layer

Model artifacts and farm parameter storage

Backend Architecture

FastAPI Framework

High-speed Python REST API implementing security best practices and handling all machine learning inference tasks

LLM Integration

Secure interactions with external LLM providers (Groq) ensuring API keys never expose to client-side

Dynamic Retraining

Automated model retraining script deployed on Render container platform



Frontend Architecture



React/Vite Foundation

Modern SPA architecture with hot module replacement and optimized bundling for rapid development cycles



Physics-Based Animations

Framer Motion delivers complex, responsive animations with 10/10 user experience scores



Vercel Edge Network

Global CDN deployment ensures sub-100ms latency worldwide through distributed edge nodes

Machine Learning Methodology

01

Data Collection

28,242 records spanning 101 countries and 10 major crops from Kaggle dataset

02

Preprocessing

Label encoding for categorical variables, standard scaling for continuous features

03

Model Selection

Ensemble methods: Random Forest and Gradient Boosting for regression tasks

04

Hyperparameter Tuning

GridSearchCV with 3-fold cross-validation optimizing `n_estimators` and `max_depth`

Evaluation Metrics

MAE

Mean Absolute Error

Average magnitude of prediction errors across all samples

RMSE

Root Mean Squared Error

Penalizes larger errors through squared loss function

R^2

R-squared Coefficient

Variance explained by the regression model



Agentic AI Workflow



Powered by LangGraph and Llama-3.3-70b via Groq for natural language understanding and structured output generation

An abstract graphic on the left side of the slide featuring stylized leaves in various shades of blue and white, set against a dark blue background. The leaves are layered and flowing, creating a sense of movement.

Future Enhancements

Granular Data Integration

Incorporate micro-climate sensors and precise soil composition measurements for field-level accuracy

RAG Implementation

Retrieval-Augmented Generation over localized agricultural extension documents to reduce LLM hallucinations

Temporal Modeling

Integrate LSTMs or RNNs to capture seasonal weather patterns and temporal dependencies